

Evaluation Metrics

1. Provenance Validity

Definition

Provenance validity measures whether each extracted entity or process step is explicitly grounded in a verifiable location within the original paper.

Implementation

For each extracted entry (targets, chemicals, synthesis steps, characterization methods), we check:

- The existence of a **prov** field
- **prov** must be a non-empty list
- At least one provenance reference contains an explicit locator such as:
 - "Page"
 - "Figure"
 - "Table"
 - "S" (Supplementary information)

Scoring

- Binary scoring per entry:
 - **1** if provenance is present and valid
 - **0** otherwise

This ensures that extracted information is traceable to the source document rather than inferred or hallucinated.

2. Units Policy Check

Purpose

The units policy check enforces consistency and normalization of scientific units across extracted numerical values.

Fixed Units Policy

Quantity	Required Unit
Temperature	K
Pressure	bar
Time	min

Mass	g
Volume	mL
Amount	mmol
Concentration	M
Capacity	mAh g ⁻¹
Yield	%

Implementation

- Units are checked in:
 - Synthesis step conditions (temperature, pressure, duration)
 - Chemical amounts
 - Final outcomes (yield, capacity)
- Any deviation from the predefined unit policy flags a violation.

Scoring

- Units policy is treated as a **hard constraint**:
 - **Pass (1)** if all units comply
 - **Fail (0)** if any violation is detected
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3. Hallucination Detection

Method

We compare the extracted JSON against the original paper text (PDF-extracted or pre-saved text) and flag suspicious cases when:

- Chemical names do not appear in the paper text
- Target compound names do not appear in the paper text
- Synthesis step descriptions do not appear in the paper text
- Characterization methods do not appear in the paper text (excluding common abbreviations such as SEM, XRD, TEM, etc.)

Threshold

- If more than **3 suspicious items** are detected → hallucination check fails
- Otherwise → hallucination check passes

Scoring

- Binary:
 - **1** = hallucination check passed

- **0** = failed

This provides a conservative, text-grounded hallucination filter without relying on LLM-based judges.

4. Overall Scoring Scheme

Scoring Structure

- Field-level completeness (schema, metadata, synthesis, characterization, etc.)
- Entity-level validity:
 - Presence of required fields
 - Provenance validity
 - Ontology existence
 - Molecular formula validity (checked against the periodic table using RDKit)
- Hallucination check (1 point)
- Units policy check (1 point)

Final Score

$$\text{Score (\%)} = \frac{\text{Number of passed checks}}{\text{Total checks}} \times 100$$

Verdict Categories

- $\geq 90\%$ → Excellent
- $\geq 75\%$ → Good
- $\geq 60\%$ → Acceptable
- $< 60\%$ → Poor

In our pipeline, papers with $\geq 90\%$ **score** are considered high-quality extractions.

5. o3 vs o4-mini

Using the above metrics, we observe that:

- **o3** generally achieves higher schema completeness and handles structurally complex papers better, but tends to introduce more hallucinated entities, leading to more frequent failures in the hallucination check.
- **o4-mini** is more conservative with hallucinations but often misses schema fields, provenance, or detailed entities.

→ “o3 is better” refers to its capability on complex extraction tasks, not necessarily a higher overall accuracy under strict hallucination constraints.